species are now found to have. Second, this change was accompanied by a decrease in symmetry: before the formation of the Higgs ocean, all particles had the same mass—zero—a highly symmetric state of affairs. If you were to exchange one particle species' mass with another, no one would know, because the masses were all the same. But after the Higgs field condensed, the particle masses transmuted into nonzero—and nonequal— values, and so the symmetry between the masses was lost.

In fact, the reduction in symmetry arising from the formation of the Higgs ocean is more extensive still. Above 10^{15} degrees, when the Higgs field had yet to condense, not only were all species of fundamental matter particles massless, but also, without the resistive drag from a Higgs ocean, all species of force particles were massless as well. (Today, the W and Z messenger particles of the weak nuclear force have masses that are about 86 and 97 times the mass of the proton.) And, as originally discovered in the 1960s by Sheldon Glashow, Steven Weinberg, and Abdus Salam, the masslessness of all the force particles was accompanied by another, fantastically beautiful symmetry.

In the late 1800s Maxwell realized that electricity and magnetism, although once thought to be two completely separate forces, are actually different facets of the same force—the electromagnetic force (see Chapter 3). His work showed that electricity and magnetism complete each other; they are the yin and yang of a more symmetric, unified whole. Glashow, Salam, and Weinberg discovered the next chapter in this story of unification. They realized that before the Higgs ocean formed, not only did all the force particles have identical masses—zero—but the photons and W and Z particles were identical in essentially every other way as well.[10] Just as a snowflake is unaffected by the particular rotations that interchange the locations of its tips, physical processes in the absence of the Higgs ocean would have been unaffected by particular interchanges of electromagnetic and weak-nuclear-force particles—by particular interchanges of photons and W and Z particles. And just as the insensitivity of a snowflake to being rotated reflects a symmetry (rotational symmetry), the insensitivity to interchange of these force particles also reflects a symmetry, one that for technical reasons is called a *gauge symmetry.* It has a profound implication. Since these particles convey their respective forces—they are their force's messenger